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PhyGuard: Physics-guided machine learning for selective diagnosis of multivariate OFDM link anomalies

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ABSTRACT

Wireless monitoring can identify anomalous intervals in OFDM links, but an alarm alone cannot distinguish interference, blockage, rapid channel variation, delayed link adaptation, and non-physical throughput disturbances. This paper presents PhyGuard, a domain-guided selective-learning framework for trustworthy diagnosis of multivariate OFDM telemetry. It combines physics-consistency evidence with a learned evidence-sufficiency gate and mechanism resolver, abstaining when evidence does not support a closed-set decision. On a 1,080-sequence waveform-grounded benchmark, five repeated tests yield 80.44% macro-F1, 76.81% physical-event coverage, 93.13% selected accuracy, and a 5.28% false-specific rate, providing the highest observed macro-F1 and coverage among evaluated selective tabular methods. An independently implemented, protocol-frozen Sionna-based evaluation contains 840 planned 1×2 SIMO CDL-A/B/C sequences; 825 satisfy prespecified quality criteria. Without Sionna-side fitting or threshold recalibration, the frozen pipeline achieves 60.41% selected accuracy, 39.83% exact physical diagnosis, and 90.08% control abstention, indicating strong control rejection alongside mechanism-dependent transfer. In an aligned comparison with 14 temporal classifiers, no comparator Pareto-dominates PhyGuard on the prespecified mean five-metric profile; among models covering over 80% of physical events, PhyGuard has the lowest false-specific rate. Detector, sensitivity, ablation, and complexity analyses further characterize the operating range. The results support an auditable, risk-controlled link-level diagnostic framework.

Glossary

Candidate interval: a time interval proposed by an anomaly detector for mechanism-level analysis.

Physical-event coverage: the fraction of true physical anomalies for which a specific diagnosis is emitted.

False-specific rate: the fraction of normal or non-physical controls that receive a physical mechanism label.

Selected accuracy: accuracy among physical anomalies for which the system emits a mechanism label.

Selective risk: one minus accuracy at a chosen retained coverage after ranking cases by confidence.

1. Introduction

Orthogonal frequency-division multiplexing (OFDM) is central to contemporary broadband wireless systems because it converts a frequency-selective channel into parallel subchannels and supports flexible link adaptation [1,2]. Its operational performance, however, can deteriorate for physically different reasons. Co-channel interference raises received energy while reducing signal quality; blockage attenuates the desired

path; rapid channel variation degrades tracking and equalization; and delayed modulation-and-coding adaptation produces excessive decoding errors after a channel change. These mechanisms can lead to similar user-visible symptoms, such as reduced goodput, even though the appropriate mitigation is different.

Modern radio equipment exposes a growing collection of physical- and link-layer key performance indicators (KPIs), including received signal power, signal-to-interference-plus-noise ratio (SINR), error vector magnitude (EVM), channel-quality indicator (CQI), modulation and coding scheme (MCS), block or packet error statistics, and retransmission feedback. Such telemetry has enabled machine-learning methods for link-quality estimation and anomaly detection [3,4]. Detection is nevertheless only the first step. A detector answers whether an interval is unusual; it does not necessarily provide a defensible explanation of the physical mechanism.

Mechanism attribution is particularly unsafe when a model is forced to classify every alarm. A throughput decrease with stable radio indicators may originate above the physical layer. A mild fading event can produce evidence too weak to distinguish blockage from normal shadowing. An inaccurate detector may select an interval that only partially overlaps

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the event. In all three cases, a confident physical diagnosis can be more harmful than an abstention because it may trigger an inappropriate interference, mobility, or adaptation response.

Selective classification formalizes the option to reject uncertain cases [5–7]. A selective system jointly controls coverage and risk instead of maximizing unconditional classification accuracy. Existing reject-option methods are largely domain independent, whereas wireless physical-layer diagnosis offers useful structure: the plausibility of a mechanism can be checked against relations among received power, aggregate energy, signal quality, equalization error, link adaptation, and delivered goodput. For example, interference is more plausible when SINR and EVM degrade without a commensurate loss of desired-signal power; blockage is more plausible when desired power and SINR fall together; and adaptation mismatch is characterized by a gap between the MCS in use and the MCS supported by the observed SINR.

This paper introduces PhyGuard, a lightweight domain- and physics-guided selective-learning framework for mechanism diagnosis from multivariate OFDM telemetry. The method is placed downstream of an arbitrary interval detector. It transforms a candidate interval into robust changes relative to a normal prefix, builds explicit physical-consistency scores, and separates two learned decisions: whether the available evidence is sufficient for a physical explanation and which mechanism is most plausible. Thresholds are fixed on validation data under reliability constraints, so the test set is not used to choose the operating point.

To avoid evaluating on manually specified key-performance-indicator trajectories, this study derives all telemetry from received symbols generated by a waveform-level link simulation. Random QAM symbols pass through time-varying multipath channels whose tap delays and powers follow 3GPP tapped-delay-line (TDL)-A/B/C profiles [8]. An independent OFDM interferer, channel-estimation delay, additive noise, and a compact link-adaptation controller generate telemetry from demodulation outcomes rather than from hand-written KPI trajectories. The simulator is intentionally smaller than a full 5G NR stack, but it preserves the signal path required to compute EVM, bit error rate (BER), packet error rate (PER), CQI, MCS, and goodput from received data.

The principal contributions are:

- A selective post-alarm diagnosis formulation in which normal and non-physical controls explicitly measure unsupported physical specificity rather than being forced into a closed set.
- A physics-guided evidence representation for interference, blockage, mobility-induced tracking degradation, and link-adaptation mismatch, coupled with a learned evidence-sufficiency gate and mechanism resolver.
- A validation-constrained abstention procedure that selects a coverage-risk operating point under explicit false-specific and selected-accuracy requirements.
- A 1,080-sequence source benchmark with repeated and regime-disjoint evaluations that reports physical-event coverage, macro-F1, selected accuracy, and false-specific risk jointly.
- An independently implemented, protocol-frozen Sionna-based evaluation and three-detector interval-localization study that quantify cross-generator transfer and the detector–diagnoser interface under the evaluated link-level settings without Sionna-side model fitting.
- A comprehensive robustness suite comprising 14 modern temporal classifiers, paired cluster-bootstrap uncertainty, physics-formula and policy sweeps, cross-system directional reliability, per-indicator ablation, and empirical complexity scaling.

The remainder of the paper is organized as follows. Section 2 reviews wireless anomaly diagnosis and selective prediction. Section 3 defines the waveform and diagnosis problem. Section 4 presents PhyGuard. Section 5 describes the simulator and protocols. Section 6 reports the results. Section 7 discusses scope and limitations, and Section 8 concludes the paper.

2. Related work

2.1. Wireless anomaly detection and link-quality learning

Anomaly detection methods range from statistical rules to shallow ensembles and deep representations [9–11]. Wireless links add dependencies that are absent in generic time-series benchmarks: propagation affects power and channel coherence; interference changes aggregate energy and SINR; adaptation maps estimated quality to an MCS; and decoding outcomes determine useful throughput. Reviews of machine-learning-based link-quality estimation emphasize that feature choice, environment shift, and validation design materially affect conclusions [3].

MOJO combined distributed physical-layer observations to identify 802.11 anomalies, illustrating the benefit of cross-node and cross-metric evidence [12]. Cerar et al. constructed anomalous IoT links and evaluated supervised and unsupervised detectors on empirical traces [4]. At network-management level, adaptive KPI monitoring detects abrupt performance changes [13], while XMLAD combines explainable models with mobile-network anomaly classification [14]. These studies establish the value of structured telemetry, but most focus on detection or assume that every candidate belongs to a known abnormal class.

Physical-layer interference recognition has also been studied with time-frequency images and learned classifiers [15]. Bendecheche et al. examined how intentional interference changes the behavior of machine-learning multiple-input multiple-output (MIMO) detectors [16]. The present work differs by considering multiple physical mechanisms, explicit controls that should not receive a physical label, and a selective mechanism decision rather than a forced closed-set classification.

Recent communication research also illustrates the expanding role of artificial intelligence (AI) beyond anomaly detection. Learning-driven resource scheduling has been investigated for secure multi-reconfigurable-intelligent-surface (multi-RIS)-assisted orthogonal-time-frequency-space (OTFS) transmission and for scalable space-air-ground integrated networks (SAGINs) [17–19]. At the physical layer, ReRpGAN uses generative-adversarial learning to stabilize channel estimation with one-bit analog-to-digital converters [20]. These studies address resource control, access, or channel reconstruction. In contrast, [PhyGuard](#) addresses post-alarm mechanism attribution from synchronized telemetry and explicitly retains abstention when the available indicators do not support a specific physical explanation.

2.2. Physical consistency and OFDM telemetry

OFDM system behavior is governed by relationships among the desired channel, interference, noise, equalization quality, and adaptation state. Standard wireless texts describe the links between SINR, modulation order, coding rate, and error probability [21,22]. 3GPP TS 38.214 specifies NR procedures for data and provides standardized MCS concepts [23]; this study uses a smaller four-level adaptation abstraction to keep the diagnostic experiment transparent. Channel dynamics are generated using TDL profiles from 3GPP TR 38.901 [8].

The central premise of [PhyGuard](#) is not that any single relation uniquely identifies a fault. Instead, several relations should agree before a specific diagnosis is emitted. Desired-power loss together with SINR loss supports blockage. SINR and EVM degradation without comparable desired-power loss supports interference. Reduced channel correlation with increased EVM supports mobility-induced tracking error. An MCS that remains above the level supported by current SINR, together with rising PER, supports adaptation mismatch. This view is related to model-driven machine learning: known signal structure is used to constrain a learned decision rather than replacing learning with a complete analytical model.

2.3. Modern temporal classification baselines

Modern time-series classification spans random convolutional transforms, competing-kernel methods, deep convolutional networks, recurrent models, and attention-based sequence models. MiniRocket and MultiRocket provide efficient random-kernel transform baselines [24,25]; HYDRA combines competing convolutional kernels with dictionary-style counting [26]. InceptionTime and fully convolutional networks provide strong convolutional reference points [27,28]. Temporal convolutional networks (TCNs), long short-term memory (LSTM), and Transformer encoders represent dilated convolutional, recurrent, and attention-based sequence modeling [29–31]. We evaluate these families under frozen split memberships and validation-only selective-output rules, providing a substantially broader comparison than shallow tabular baselines alone.

2.4. Selective prediction

A reject option permits a classifier to trade coverage for reliability [5,6]. SelectiveNet integrates prediction and selection in a neural architecture [7], but the same operational principle applies to shallow models when confidence thresholds are calibrated independently. Existing selective classifiers primarily derive rejection from predictive confidence, whereas generic temporal classifiers do not explicitly assess whether a class-specific explanation is consistent with domain relations. PhyGuard addresses this gap by separating evidence sufficiency from mechanism attribution. In wireless diagnosis, generic confidence alone is insufficient because a model can be confident for a physically contradictory pattern. We therefore use a physical-validity gate and a mechanism threshold, and explicitly evaluate false-specific diagnoses on controls.

3. System model and problem definition

3.1. Waveform model

We consider a single-input single-output cyclic-prefix OFDM (CP-OFDM) link with $N = 64$ subcarriers, of which 52 are active. For frame t , OFDM symbol m , and active subcarrier k , the received frequency-domain sample is

$$Y_{t,m,k} = \sqrt{P_{s,i}} H_{t,k} X_{t,m,k} + \sqrt{P_{i,i}} G_{t,k} I_{t,m,k} + N_{t,m,k}, \quad (1)$$

where $X_{t,m,k}$ is the desired QAM symbol, $I_{t,m,k}$ is an independent co-channel OFDM interferer, $H_{t,k}$ and $G_{t,k}$ are desired and interfering channel responses, and $N_{t,m,k} \sim \mathcal{CN}(0, 1)$. The desired response is

$$H_{t,k} = \sum_{\ell=1}^L h_{t,\ell} \exp(-j2\pi f_k \tau_{\ell}), \quad (2)$$

with delays τ_{ℓ} and average powers taken from the selected TDL profile. The tap evolution is approximated by

$$h_{t,\ell} = \rho h_{t-1,\ell} + \sqrt{1 - \rho^2} u_{t,\ell}, \quad \rho = J_0(2\pi f_D T_f), \quad (3)$$

where $u_{t,\ell}$ has the profile tap variance, f_D is Doppler frequency, $T_f = 1$ ms, and $J_0(\cdot)$ is the zeroth-order Bessel function.

The receiver uses a one-frame delayed channel estimate with estimation noise and a low-pass update. This compact model introduces realistic tracking loss as Doppler increases. Equalized symbols are hard demapped to recover bits. EVM, BER, and PER are computed from the transmitted and detected symbols. The simulator does not implement channel coding or hybrid automatic repeat request (HARQ) scheduling; PER is obtained from 48-bit packet surrogates and is used as a reliability indicator.

3.2. Telemetry sequence

Each sequence is $\mathbf{x}_{1:T} \in \mathbb{R}^{T \times 10}$ with $T = 80$ frames and the following indicators:

$$\mathbf{x}_t = [\text{RSRP}, \text{RSSI}, \text{SINR}, \text{EVM}, \text{BER}, \text{PER}, \text{CQI}, \text{MCS}, \text{goodput}, \rho_H]_t, \quad (4)$$

where ρ_H is normalized correlation between consecutive channel responses. Reference-signal received power (RSRP) is represented by desired received power, received signal strength indicator (RSSI) by desired plus interference and noise power, and goodput by correctly received information bits normalized by the maximum rate in the compact MCS table.

An upstream detector proposes an interval $[a, b)$. The diagnosis target is

$$y \in \mathcal{Y} = \{\text{interference, blockage, mobility, adaptation mismatch}\}. \quad (5)$$

Normal sequences and non-physical goodput degradation are controls rather than additional physical mechanisms. The output is $\hat{y} \in \mathcal{Y} \cup \{\perp\}$, where $\perp$ denotes abstention.

The present task is deliberately restricted to one dominant mechanism per candidate interval. Simultaneous blockage, interference, hardware impairment, and traffic-layer disturbance are not interpreted as independent labels that can be activated together. A mixed event may be mapped to the dominant learned class or rejected by the gate; multi-label co-fault diagnosis is outside the current claim and is treated as a primary limitation.

3.3. Selective objectives

Let $\mathcal{P}$ be physical events and $\mathcal{C}$ be controls. With $e_i = 1$ when $\hat{y}_i \neq \perp$, selected accuracy, physical coverage, and false-specific rate are

$$\text{SelAcc} = \frac{\sum_{i \in \mathcal{P}} e_i \mathbb{1}[\hat{y}_i = y_i]}{\sum_{i \in \mathcal{P}} e_i}, \quad (6)$$

$$\text{Cov} = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} e_i, \quad (7)$$

$$\text{FSR} = \frac{1}{|\mathcal{C}|} \sum_{i \in \mathcal{C}} e_i. \quad (8)$$

Macro-F1 is computed over the four physical labels, with abstentions counted as missed physical events. This prevents a method from obtaining an attractive selected accuracy by emitting only a very small number of decisions.

4. Proposed PhyGuard framework

Fig. 1 summarizes the end-to-end pipeline. The design is deliberately modular: the upstream interval detector and the downstream diagnostic resolver can be developed and calibrated independently.

4.1. Reference-calibrated raw features

The first 24 frames are treated as a normal reference prefix. For KPI j , its reference median and robust scale are

$$\mu_j = \text{median}_{t \leq 24} x_{t,j}, \quad s_j = 1.4826 \text{median}_{t \leq 24} |x_{t,j} - \mu_j| + \epsilon, \quad (9)$$

following the robust median absolute deviation principle [32]. Candidate samples are standardized as $z_{t,j} = \text{clip}[(x_{t,j} - \mu_j)/s_j, -10, 10]$. For every KPI, six interval summaries are extracted: mean, standard deviation, median, 10th percentile, 90th percentile, and linear slope. The resulting raw vector $\mathbf{r} \in \mathbb{R}^{60}$ is intentionally model agnostic.

4.2. Physical-consistency evidence

We additionally compute interpretable changes between candidate and reference means. The evidence includes desired-power drop, RSSI rise, SINR drop, EVM/BER/PER rise, CQI and MCS drop, goodput drop, channel-correlation drop, the change in the RSSI-RSRP gap, and an adaptation gap between the used MCS and the MCS supported by current SINR.

These changes are combined into four soft mechanism scores. Denoting positive part by $[q]_+ = \max(q, 0)$ and a stability term by $\phi(q; s) = \exp(-|q|/s)$, the scores follow the templates

$$p_{\text{int}} = \frac{1}{5} ([\Delta_{\text{gap}}]_+ + [\Delta_{\text{SINR}}]_+ + \phi(\Delta_{\text{RSRP}}) + [\Delta_{\text{EVM}}]_+ + [\Delta_{\text{BER}}]_+), \quad (10)$$

PhyGuard: waveform-grounded selective diagnosis

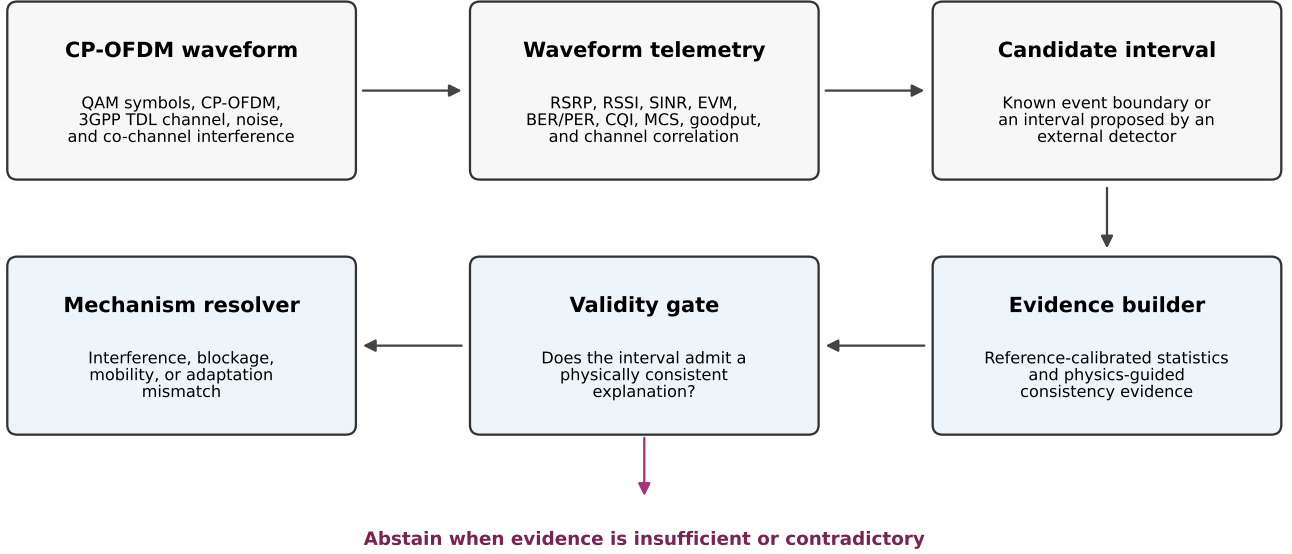


Fig. 1. Post-detection PhyGuard workflow. An external detector (or the known event boundary in the source-domain diagnostic study) defines the candidate interval. The evidence builder extracts 60 reference-calibrated raw statistics and 19 physics-consistency features. The validity gate scores evidence sufficiency from the physics-consistency vector, and the mechanism resolver consumes the combined 79-dimensional feature vector to emit one dominant mechanism label only when both decision thresholds are satisfied; otherwise PhyGuard abstains. Section 6.8 replaces known boundaries with three frozen detector proposals to assess upstream-interval sensitivity.

Algorithm 1 PhyGuard training and selective inference.

Require: Training, validation, and test candidate intervals

- 1: **for** each sequence **do**
- 2: Build raw interval statistics $\mathbf{r}$ and physical evidence $\mathbf{p}$
- 3: **end for**
- 4: Fit validity gate on all training controls and physical events
- 5: Fit mechanism resolver on training physical events only
- 6: Choose (τ_g, τ_c) on validation data using Eq. (15)
- 7: **for** each test candidate **do**
- 8: Compute g and mechanism probabilities q_k
- 9: Emit Eq. (14); otherwise return abstention
- 10: **end for**

$$p_{\text{blk}} = \frac{1}{5} ([\Delta_{\text{RSRP}}]_+ + [\Delta_{\text{SINR}}]_+ + [\Delta_{\text{EVM}}]_+ + [\Delta_{\text{CQI}}]_+ + \phi(\Delta_{\text{gap}})), \quad (11)$$

$$p_{\text{mob}} = \frac{1}{5} ([\Delta_{\rho_H}]_+ + [\Delta_{\text{EVM}}]_+ + [\Delta_{\text{BER}}]_+ + \phi(\Delta_{\text{RSRP}}) + \phi(\Delta_{\text{SINR}})), \quad (12)$$

$$p_{\text{mis}} = \frac{1}{5} ([\Delta_{\text{adapt}}]_+ + [\Delta_{\text{SINR}}]_+ + [\Delta_{\text{EVM}}]_+ + [\Delta_{\text{BER}}]_+ + [\Delta_{\text{CQI}}]_+ - 0.35[\Delta_{\text{MCS}}]_+). \quad (13)$$

A fifth score represents non-physical goodput degradation: goodput falls while radio power, quality, EVM, and BER remain stable. The complete physical vector $\mathbf{p} \in \mathbb{R}^{19}$ contains these scores, their maximum and margin, and the underlying normalized changes. It is not a hard rule system; it exposes communication structure to the learned models.

Meaning of “physics-guided”. Here, “physics-guided” does not mean a physics-informed neural network, a differentiable simulator, or a physics-constrained optimization objective. Instead, the guidance is representational and architectural: communication-domain directional relations are encoded into an explicit, auditable evidence vector, which is then processed by a learned logistic validity gate and an Extra-Trees mechanism resolver. From a machine-learning perspective, PhyGuard is a selective multivariate time-series classifier with an explicit evidence-sufficiency gate, a domain-guided representation, and a reject option for controlling unsupported specific predictions. The downstream models remain data-

driven; physics structures the evidence and diagnostic hypothesis space rather than replacing learning with hand-crafted rules.

Role and boundary of PER. In the compact source simulator, PER is computed from 48-bit uncoded packet surrogates. It is therefore used only as an internal reliability proxy, not as a calibrated estimate of coded NR packet-error or block-error probability. A PER increase is retained among the underlying normalized changes, but PER is not inserted directly into Eqs. (10)–(13). Because the learned models may still exploit the underlying PER feature, any PER-based interpretation in this study is restricted to the source benchmark and would require coded-link validation before operational deployment (Table 1).

4.3. Two-stage selective resolver

A logistic-regression gate estimates $g(\mathbf{p}) = P(\text{physical event} | \mathbf{p})$. An extremely randomized trees classifier [33] estimates mechanism probabilities $q_k(\mathbf{r}, \mathbf{p})$ for $k \in \mathcal{Y}$, using only physical events for training. The final decision is

$$\hat{y} = \begin{cases} \arg \max_k q_k, & g \geq \tau_g \text{ and } \max_k q_k \geq \tau_c, \\ \perp, & \text{otherwise.} \end{cases} \quad (14)$$

The separation prevents the mechanism classifier from learning an artificial “other” class that mixes normal operation with non-physical degradation.

4.4. Validation-constrained operating point

Thresholds (τ_g, τ_c) are selected on validation data by maximizing physical coverage under two target constraints:

$$\max_{\tau_g, \tau_c} \text{Cov}_{\text{val}} \quad \text{s.t.} \quad \text{FSR}_{\text{val}} \leq 0.05, \quad \text{SelAcc}_{\text{val}} \geq 0.90. \quad (15)$$

If no grid point satisfies both constraints, a deterministic penalty ranks the violations before coverage. Thresholds are never adjusted on test results. Algorithm 1 presents the complete procedure.

Table 1

Physical evidence used by the four mechanism hypotheses. Arrows denote expected candidate changes relative to the reference prefix.

Mechanism	RSRP	RSSI-RSRP	SINR	EVM/BER	ρ_H	MCS gap
Interference	stable	↑	↓	↑	variable	small
Blockage	↓	stable	↓	↑	variable	adaptive
Mobility	stable	stable	near stable	↑	↓	small
Adaptation mismatch	variable	stable	↓	↑	variable	↑

Table 2

Common waveform and sequence parameters.

Parameter	Value
FFT size / active subcarriers	64 / 52
Subcarrier spacing	30 kHz
OFDM symbols per frame	4
Frame duration / sequence length	1 ms / 80 frames
Reference prefix	first 24 frames
Event start / duration	frames 30–37 / 22–29 frames
Desired modulation	QPSK, 16-QAM, or 64-QAM
Interferer modulation	independent QPSK or 16-QAM
Channel profiles	3GPP TDL-A, TDL-B, TDL-C
Random seed	20260705

Table 3

Propagation regimes. SNR and baseline Doppler are sampled uniformly within the shown ranges.

Regime	TDL	RMS delay	SNR (dB)	Doppler (Hz)
Indoor	A	30 ns	16–25	1–8
Urban	B	100 ns	9–19	8–45
Highway	C	300 ns	10–21	70–180
Cell edge	C	300 ns	1–11	5–45

5. Experimental methodology

5.1. Waveform-level benchmark

The benchmark uses a compact CP-OFDM simulation implemented in Python/NumPy. OFDM itself is a well-established multicarrier technique [1]; the purpose here is not to introduce a new transceiver but to derive diagnostic telemetry from received symbols. Table 2 lists the common parameters.

The TDL normalized tap delays and powers are taken exactly from Tables 7.7.2-1–3 of 3GPP TR 38.901 version 19.2.0 [8]; delays are scaled to the regime-specific RMS delay spread. Four regimes are used (Table 3). The benchmark is balanced: each regime contains 45 sequences for each of six labels, for 270 sequences per regime and 1080 in total. Within each regime-label combination, mild, moderate, and severe settings occur 15 times each.

5.2. Fault interventions and controls

Table 4 defines the controlled interventions. The parameter values determine only data generation; severity labels are not given to the diagnostic model. An interference event introduces an independent OFDM signal at jammer-to-signal ratios of -7 , -2 , or 3 dB. Blockage attenuates the desired signal by 4, 8, or 12 dB. Mobility raises Doppler to 180, 420, or 850 Hz, stressing the delayed channel estimate. Adaptation mismatch reduces link quality by 4, 8, or 12 dB while holding the previous MCS for 5, 10, or 16 frames. The non-physical control caps delivered goodput without modifying the received waveform; normal controls contain no intervention.

Fig. 2 shows representative standardized trajectories. The mechanisms are not separated by a single perfect indicator. For instance, both blockage and adaptation mismatch can reduce SINR, while mobility mainly appears through EVM, BER, and channel-correlation changes.

Table 4

Mechanism interventions at mild, moderate, and severe levels.

Condition	Mild	Moderate	Severe
Interference J/S	-7 dB	-2 dB	3 dB
Blockage attenuation	4 dB	8 dB	12 dB
Mobility Doppler	180 Hz	420 Hz	850 Hz
Adaptation quality loss	4 dB	8 dB	12 dB
MCS hold duration	5 frames	10 frames	16 frames
Non-physical goodput cap	0.72	0.48	0.25

5.3. Comparison methods

The following methods use the same train, validation, and test indices.

- **Single-KPI rules:** the largest of four physical scores is emitted when a fixed evidence requirement is met.
- **Logistic regression:** standardized raw statistics with class balancing and confidence rejection.
- **Random forest:** 300 trees, maximum depth 12, and class-balanced training [34].
- **Gradient boosting:** histogram gradient boosting with 220 iterations; gradient boosting follows the additive tree principle introduced by Friedman [35].
- **Two-stage without physics:** the same gate-resolver structure as PhyGuard, but both stages use only raw interval statistics.
- **PhyGuard without abstention:** the complete feature set, but the highest-probability mechanism is always emitted.
- **PhyGuard:** the complete proposed system.

Logistic regression, forests, PCA, and Isolation Forest are implemented using scikit-learn [36]. Hyperparameters are fixed before test evaluation; only confidence thresholds are selected on validation data.

5.4. Evaluation protocols

Repeated main evaluation. Five repetitions use stratified 60/20/20 train/validation/test splits. Stratification is by regime, label, and severity. Each test split contains 216 sequences: 144 physical events and 72 controls. Ground-truth event boundaries are used to isolate mechanism diagnosis from localization.

Unseen-regime evaluation. Each of the four regimes is held out in turn. The other three regimes are split into training and validation subsets, while all 270 sequences from the held-out regime form the test set. This protocol evaluates propagation and operating-condition shift.

Source-domain detector evaluation. Ground-truth boundaries are replaced by 24-frame intervals proposed by three unsupervised detectors: robust multivariate energy based on median absolute deviations, principal component analysis (PCA) reconstruction error [37], and Isolation Forest [38]. This source-domain experiment provides an in-distribution diagnostic check, while the independently implemented Sionna-based protocol below supplies the principal detector-sensitivity evidence under implementation and channel shift.

Risk-coverage evaluation. Physical test events are ranked by the product of gate and mechanism confidence. Risk is the fraction of incorrect or abstained cases within the retained top-confidence subset. This ranking is distinct from the fixed deployment threshold and shows how reliability changes with retained coverage.

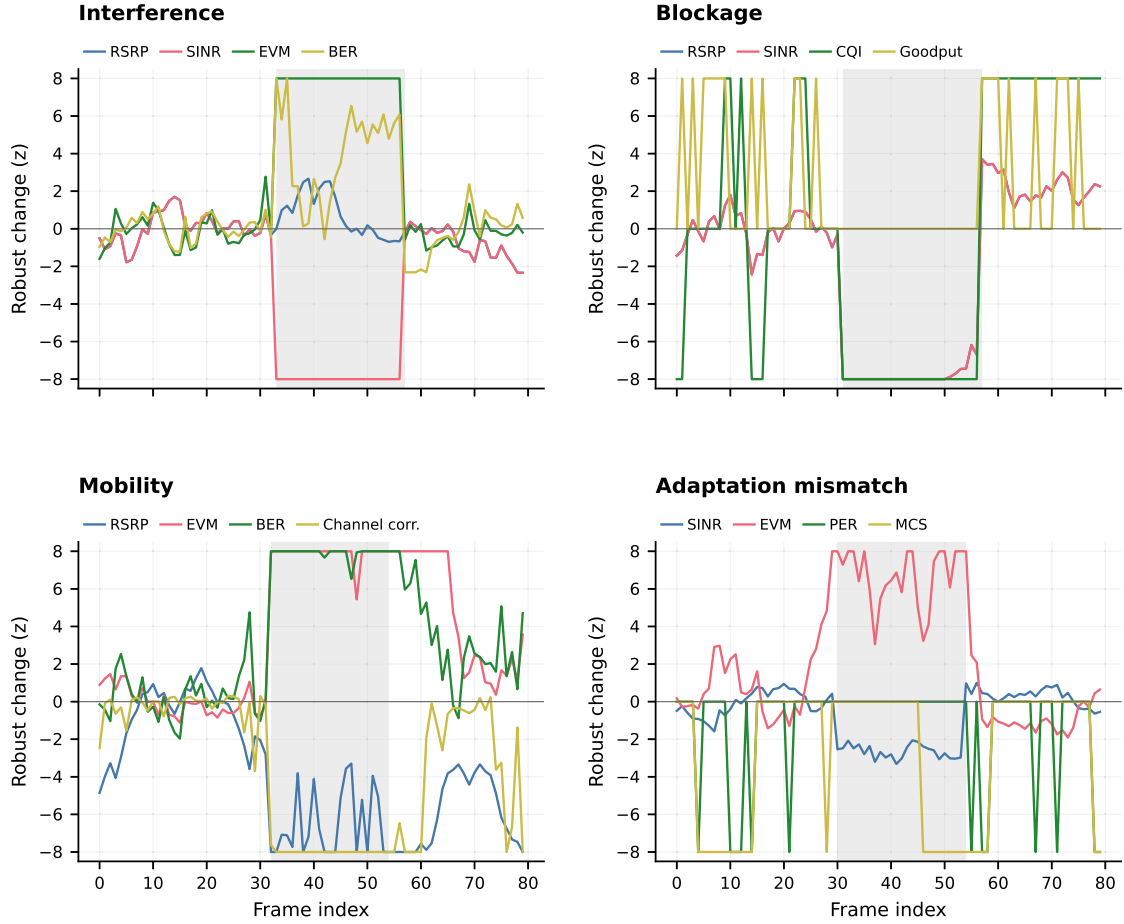


Fig. 2. Representative severe urban events. Each trajectory is robustly standardized relative to its normal prefix; the shaded region is the injected event. The plotted indicators are selected for readability, whereas the model receives all ten KPIs.

5.5. Independently implemented Sionna-based link-level pressure test

To test transfer beyond the compact NumPy simulator, we constructed a separately implemented link-level benchmark with Sionna [39]. The protocol was frozen before formal generation and contains

$$3 \text{ CDL models} \times (4 \text{ mechanisms} \times 3 \text{ severities} + 2 \text{ controls}) \times 20 = 840$$

planned sequences. The channels use CDL-A, CDL-B, and CDL-C models in a 1×2 SIMO configuration. The four interventions are interference, blockage, mobility with fixed two-frame channel-state aging, and adaptation mismatch; normal and non-physical goodput sequences serve as controls. The locked quality-control pipeline retained 825 sequences (706 physical anomalies and 119 controls): one sequence failed the artifact-quality check and 14 mild interventions did not satisfy the pre-specified mechanism-direction criterion. The quality rules were applied before downstream inference, and no excluded sample was regenerated or replaced. The reported analysis follows this prespecified protocol and covers the 825 sequences satisfying both artifact-quality and mechanism-direction criteria; it therefore characterizes protocol-compliant transfer rather than unconditional performance over all generated interventions.

For reproducible formal evaluation, we instantiated a locked reconstruction of the manuscript-defined pipeline. The 60 raw statistics, 19 physics features, 79-dimensional resolver input, class order, five gate-resolver bundles, and five validation-derived operating points were fixed before Sionna inference. No model fitting, feature selection, threshold calibration, or detector selection used the 825-sequence formal set. Primary transfer results condition diagnosis on the locked candidate interval.

Uncertainty is estimated by 10,000 sample-cluster bootstrap resamples, keeping the five frozen model repeats together.

5.6. Frozen upstream-detector sensitivity protocol

The same 825 sequences are subsequently evaluated with detector-proposed intervals. Robust Energy requires no fitting. PCA is fitted independently to the first 24 reference frames of each sequence. Isolation Forest is fitted once, before formal inference, on 25,920 standardized reference frames from the source-domain benchmark and is never fitted on Sionna data. Each detector selects the earliest 24-frame window with maximum mean score after the reference prefix. All three detector algorithms, fitting rules, random seeds, and window settings were frozen before proposal generation. Localization is evaluated on the 706 physical events, whereas downstream selective diagnosis is evaluated on all 825 sequences. Detector-conditioned predictions are paired sample-by-sample and repeat-by-repeat with the locked-interval predictions.

5.7. Modern temporal-baseline protocol

The extended benchmark includes 14 temporal classifiers spanning random-kernel transforms, competing-kernel methods, one-dimensional convolutional networks, recurrent networks, and a compact Transformer. All models receive the same 80×10 sequence tensor and use the same five frozen 648/216/216 train/validation/test memberships. The temporal models solve a six-class task containing the four physical mechanisms and two controls. Predictions of either control class map to abstention; a physical class is emitted only when its maximum physical-class score

Table 5

Repeated main evaluation (mean $\pm$ standard deviation over five stratified splits). Higher is better for selected accuracy, coverage, macro-F1, and balanced accuracy; lower is better for false-specific rate.

Method	Selected accuracy (%)	Physical coverage (%)	False-specific rate (%)	Macro-F1 (%)	Balanced accuracy (%)
Single-KPI rules	69.11 $\pm$ 1.44	62.50 $\pm$ 2.25	15.00 $\pm$ 2.06	51.86 $\pm$ 1.05	43.19 $\pm$ 1.80
Logistic regression	97.28 $\pm$ 2.51	61.53 $\pm$ 7.34	4.17 $\pm$ 3.54	70.50 $\pm$ 4.46	59.72 $\pm$ 5.62
Random forest	95.20 $\pm$ 1.84	71.67 $\pm$ 4.90	5.83 $\pm$ 1.52	78.25 $\pm$ 3.24	68.19 $\pm$ 4.26
Gradient boosting	94.73 $\pm$ 1.95	72.36 $\pm$ 4.80	6.67 $\pm$ 3.73	78.20 $\pm$ 2.25	68.47 $\pm$ 3.24
Two-stage without physics	92.82 $\pm$ 3.33	62.78 $\pm$ 4.21	6.39 $\pm$ 3.20	70.03 $\pm$ 2.82	58.19 $\pm$ 3.04
PhyGuard without abstention	86.67 $\pm$ 1.03	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	86.73 $\pm$ 0.99	86.67 $\pm$ 1.03
PhyGuard	93.13 $\pm$ 2.18	76.81 $\pm$ 1.67	5.28 $\pm$ 3.46	80.44 $\pm$ 2.20	71.53 $\pm$ 2.25

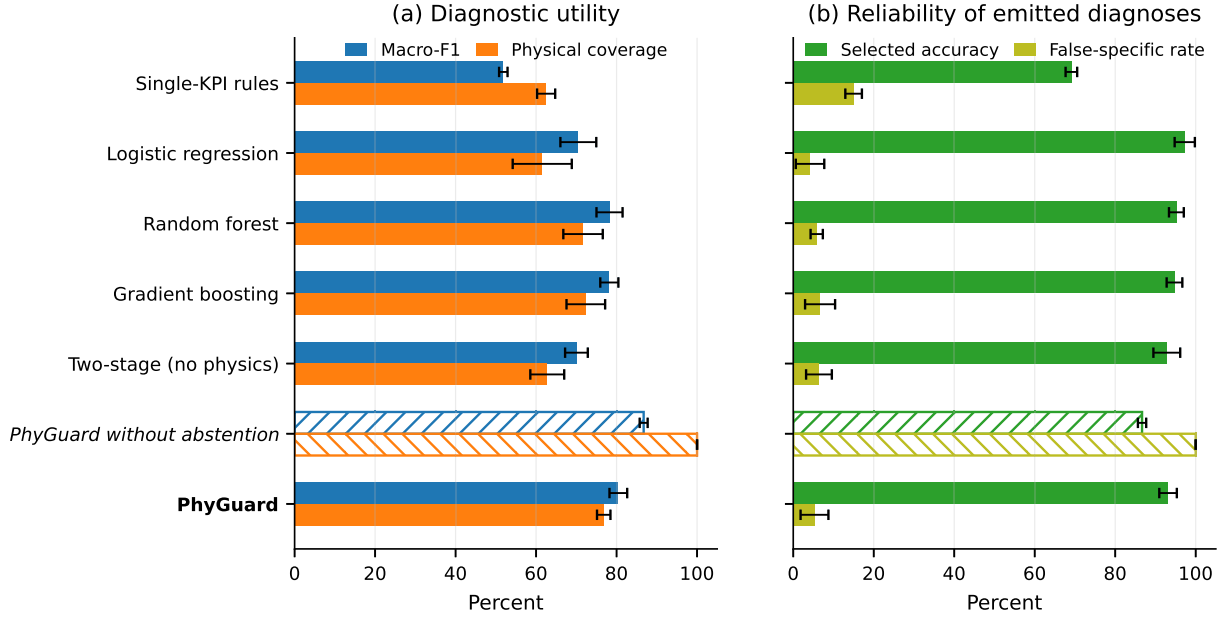


Fig. 3. Main repeated evaluation. Each split contains 216 test sequences (144 physical events and 72 controls); error bars show one standard deviation over five splits. The forced-output variant is shown with a hatched outline; its 100% false-specific rate means that every control receives a physical mechanism label.

exceeds a validation-selected threshold. The temporal threshold rule was prespecified and frozen before formal model execution (validation FSR ≤ 0.10 , selected accuracy ≥ 0.80 when feasible), while PhyGuard retains its independently frozen source operating points. Test data are not used for threshold or checkpoint selection.

The complete execution contains 14 models and 70 model-repeat tasks, covering random-kernel, competing-kernel, convolutional, recurrent, and attention-based families. Statistical comparisons use 10,000 paired cluster-bootstrap resamples over the 726 unique source-sequence indices appearing across the repeated test sets; all predictions from the same original sequence are retained in one bootstrap cluster. Holm adjustment is applied separately within each metric across the 14 comparisons.

For the temporal challenge, PhyGuard is evaluated with the locked pipeline on exactly the same row-level split memberships used by the temporal models. Table 5 reports the original main source-study evaluation, whereas Table 11 reports the separately frozen comparator-aligned replay. The two analyses use the same feature definitions and gate-resolver architecture, but their frozen split memberships and evaluation artifacts belong to different protocols. The resulting metric differences, including the coverage difference, are therefore protocol-specific estimates rather than alternative selections from a single test run; the aligned table is used only for comparisons with the 14 temporal models.

5.8. Fixed-model physics-formula and validation-policy sensitivity

To test whether the operating profile depends on a numerically fragile hand-set constant, the five frozen gate-resolver decision functions are

held fixed while three source-defined factors are varied one at a time:

$$s \in \{1.0, 1.25, 1.5, 2.0, 2.5\},$$

$$w_a \in \{0.5, 0.75, 1.0, 1.25, 1.5\},$$

$$\gamma \in \{0, 0.175, 0.35, 0.525, 0.70\}.$$

The default values are $s = 1.5$, $w_a = 1.0$, and $\gamma = 0.35$. This produces 13 unique formula configurations and 65 fixed-model inference tasks; no model training is performed.

A separate 3×3 validation-policy grid combines three FSR limits (0.025, 0.05, and 0.10) with three selected-accuracy requirements (0.85, 0.90, and 0.95), giving 45 policy evaluations across five repeats. The common factor $1/5$ in Eqs. (10)–(13) is a score normalization; the sensitivity experiment targets factors that alter the relative internal evidence balance. Before non-default analysis, the compatibility path reproduces all 1080 frozen default predictions exactly. These tests characterize robustness and operating-profile movement, not globally optimal physics weights.

5.9. Cross-system reliability and per-indicator ablation

A protocol-locked directional analysis tests whether a central communication relation—higher SINR should correspond to lower packet/block error at comparable MCS—is retained across three separately produced assets. The source and Sionna evaluations use within-sequence MCS-stratified Spearman association. The public real-measurement evaluation uses ten files from the Video Streaming Network KPIs for O-RAN Testing dataset [40] and computes within-file partial Spearman association be-

tween downlink SINR and downlink BLER while controlling continuous MCS. The objective is directional consistency across systems rather than equality of effect magnitude.

A separate packet/block-error ablation evaluates four variants: the complete model, removal of the six raw error-indicator statistics, removal of the error-specific physics score, and removal of both paths. Source-domain models are refitted for each variant using the same frozen splits and validation policy. The corresponding source-trained variants are then applied to Sionna without Sionna-side fitting or threshold adjustment. The primary retention criterion requires selected accuracy to remain within five percentage points of the complete model.

5.10. Statistical analysis, reproducibility, and computation

The source benchmark uses deterministic random seeds. Main repeated results are reported as mean $\pm$ standard deviation over five splits. The macro-F1 contrast in Section 6 uses a paired Student-*t* 95% confidence interval over the five split-level differences and is interpreted cautiously because $n = 5$. Sionna transfer intervals use 10,000 nonparametric sample-cluster percentile bootstrap resamples (seed 20260730), with all five frozen repeat predictions for a sequence kept in the same cluster. Detector-conditioned comparisons use 10,000 paired sample-cluster bootstrap resamples (base seed 20260801). Temporal-model comparisons use 10,000 paired cluster-bootstrap resamples over 726 unique source-sequence indices; Holm adjustment is applied separately within each metric across the 14 model comparisons. Pareto statements are descriptive and use the reported mean point estimates, with higher values preferred except for false-specific rate.

For the Sionna-based evaluation, the generation plan and quality rules were frozen before formal execution. The candidate-interval protocol, source-fitted Isolation Forest, five downstream model bundles, and operating thresholds were likewise version-frozen and archived. The complete 840-sample plan, retained and excluded memberships, all detector proposals, and all downstream predictions remain traceable through frozen manifests. Quality-control decisions use generator-side artifact and direction criteria only and do not depend on PhyGuard predictions.

Feature extraction and model inference are CPU-based. The scaling audit was executed on Linux 6.8 (x86-64) with Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1, and scikit-learn 1.8.0 on a 16-logical-CPU host. OpenMP, MKL, OpenBLAS, and NumExpr thread counts were fixed to one, and timing used Python's nanosecond performance counter after warm-up. Each sequence-length condition used 200 timed observations, and each batch-size condition used 100 timed observations. Sequence-length timing includes feature construction and frozen-model inference; batched throughput measures inference over prepared evidence vectors. Upstream interval localization and offline training are excluded from the online timing figures.

6. Results

6.1. Main repeated evaluation

Table 5 and Fig. 3 summarize the five repeated tests. The forced-output PhyGuard variant obtains high macro-F1 (86.73%) but assigns a physical mechanism to every control, producing a 100% false-specific rate. This shows that conventional closed-set accuracy is unsuitable when controls are present.

Among selective methods, logistic regression has the highest selected accuracy (97.28%) and lowest mean false-specific rate (4.17%), but it covers only 61.53% of physical events and reaches 70.50% macro-F1. Random forest and gradient boosting improve coverage to 71.67% and 72.36%, respectively. **PhyGuardPhyGuard** achieves the highest physical coverage (76.81%) and highest macro-F1 (80.44%) among selective methods. Its selected accuracy is 93.13%, and 5.28% of controls receive a false physical diagnosis.

The observed mean macro-F1 advantage of **PhyGuardPhyGuard** over gradient boosting is 2.24 percentage points, with a paired Student-*t* 95% confidence interval of $[-0.33, 4.82]$ points across the five split-level differences. The interval indicates that the advantage is not statistically resolved at this sample size; nevertheless, **PhyGuardPhyGuard** simultaneously increases physical coverage by 4.45 points and reduces the false-specific rate by 1.39 points relative to gradient boosting. The combined result is a favorable observed coverage-risk profile among the evaluated selective tabular methods.

6.2. Contribution of physics and abstention

Comparing **PhyGuardPhyGuard** with the two-stage no-physics ablation shows that physical evidence increases macro-F1 by 10.41 points and coverage by 14.03 points, while selected accuracy remains similar. This is the clearest contribution of the structured representation: the gate can distinguish radio-consistent events from controls without relying only on generic interval statistics.

Removing abstention increases macro-F1 because every physical event receives a label, but it also labels every control as a physical fault. The result demonstrates why macro-F1 alone is incomplete. An operational resolver must jointly report coverage and false physical specificity.

6.3. Risk-coverage behavior

Fig. 4 shows that the highest-confidence subset is highly reliable. Risk increases as milder or more ambiguous cases are retained. This monotonic trend supports the confidence score as a useful routing variable: high-confidence diagnoses can be handled automatically, while low-confidence candidates can be sent to additional sensing, a longer observation window, or human investigation.

6.4. Mechanism and severity analysis

The emitted-diagnosis confusion matrix in Fig. 5 shows that mobility is the most separable mechanism (99.34% of emitted mobility diagnoses are correct), a pattern consistent with channel-correlation and EVM cues jointly exposing tracking loss. Interference reaches 95.57%, and blockage reaches 86.40%. Adaptation mismatch is confused mainly with blockage; 11.02% of its emitted diagnoses are blockage. Both events can contain a SINR decrease, and mild mismatch may not preserve an MCS gap long enough to dominate the interval summaries.

Severity results in Fig. 6 are operationally important. Mild events receive only 45.83% coverage and 54.33% macro-F1, whereas moderate and severe events reach 87.50%/85.42% and 97.08%/93.41%, respectively. The method therefore behaves as intended: it abstains frequently on weak physical changes rather than presenting mild-event performance as if all events were equally diagnosable.

6.5. Unseen propagation regimes

Table 6 evaluates transfer to a complete regime absent from training. **PhyGuardPhyGuard** has the highest mean selected accuracy (90.20%) and macro-F1 (71.77%) across held-out regimes. The strongest generic baseline, gradient boosting, achieves 70.17% macro-F1. The no-physics ablation falls to 62.62% and has a higher false-specific rate of 7.50%.

Fig. 7 reveals that the highway fold is the most conservative. **PhyGuardPhyGuard** emits fewer diagnoses there but maintains 97.09% selected accuracy. Cell-edge transfer is the most difficult in macro-F1 because low baseline SNR makes blockage, interference, and adaptation effects less separable. These differences justify reporting regime-disjoint evaluation rather than only random splits.

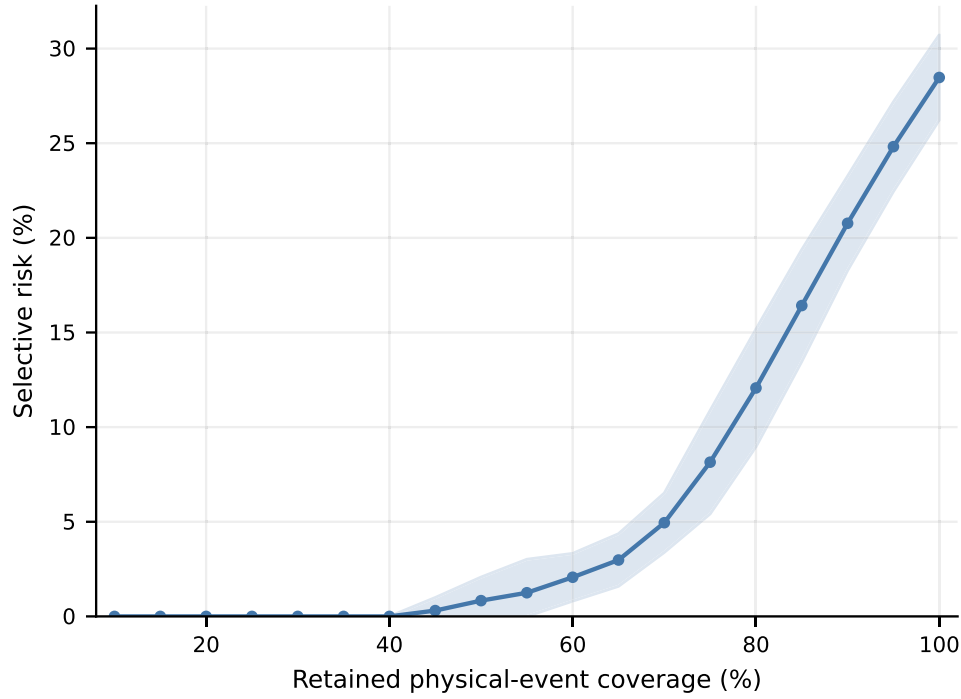


Fig. 4. Mean risk-coverage curve over five repeated test splits. Shading denotes one standard deviation. Abstentions inside a retained top-confidence subset count as errors in this ranking analysis.

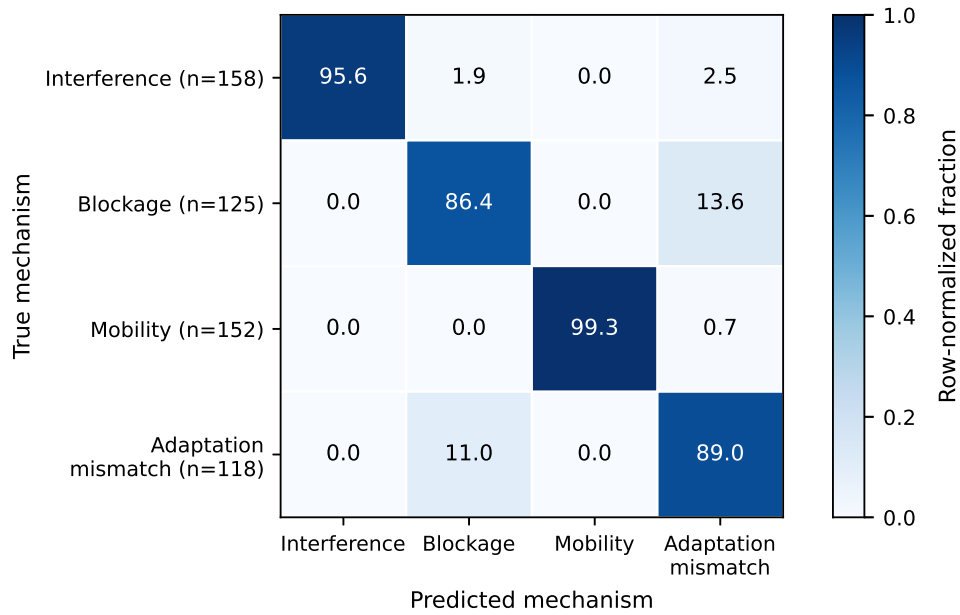


Fig. 5. Aggregate row-normalized confusion matrix over 553 emitted physical diagnoses in the five repeated tests. Row totals are 158 interference, 125 blockage, 152 mobility, and 118 adaptation-mismatch emissions. Abstentions are excluded from the matrix and reflected in coverage.

Table 6

Leave-one-regime-out results (mean $\pm$ standard deviation over four held-out regimes).

Method	Selected accuracy (%)	Coverage (%)	False-specific rate (%)	Macro-F1 (%)	Balanced accuracy (%)
Random forest	89.73 $\pm$ 6.50	66.53 $\pm$ 12.81	4.17 $\pm$ 3.67	68.16 $\pm$ 6.57	59.17 $\pm$ 8.59
Gradient boosting	87.39 $\pm$ 4.66	71.53 $\pm$ 10.81	3.89 $\pm$ 2.64	70.17 $\pm$ 7.55	62.36 $\pm$ 9.09
Two-stage without physics	88.68 $\pm$ 9.38	61.81 $\pm$ 17.94	7.50 $\pm$ 4.48	62.62 $\pm$ 13.05	53.89 $\pm$ 13.68
PhyGuard	90.20 $\pm$ 6.90	70.00 $\pm$ 10.09	4.72 $\pm$ 3.67	71.77 $\pm$ 9.00	63.06 $\pm$ 9.78

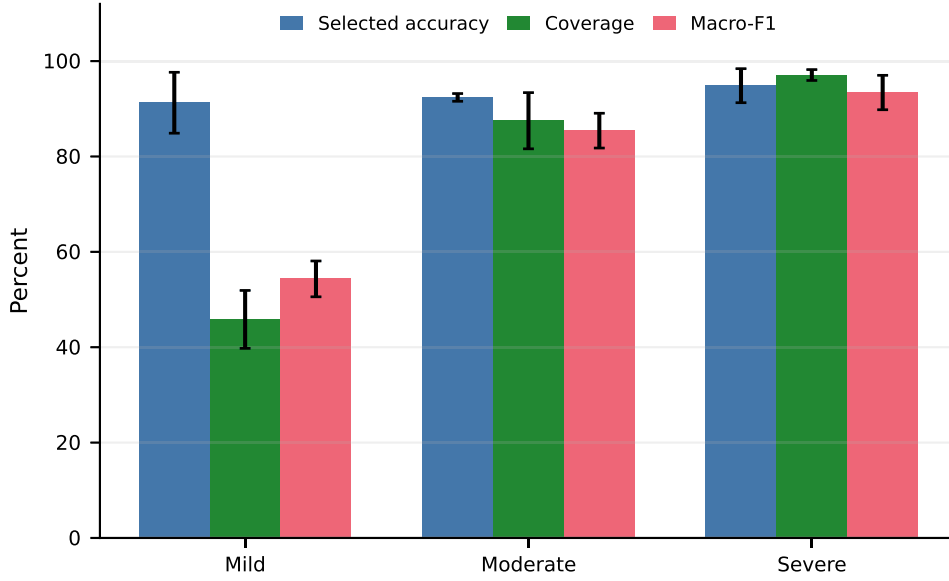


Fig. 6. PhyGuard performance by event severity. Each split contains 48 physical events per severity; bars and error bars show the mean and one standard deviation over five splits.

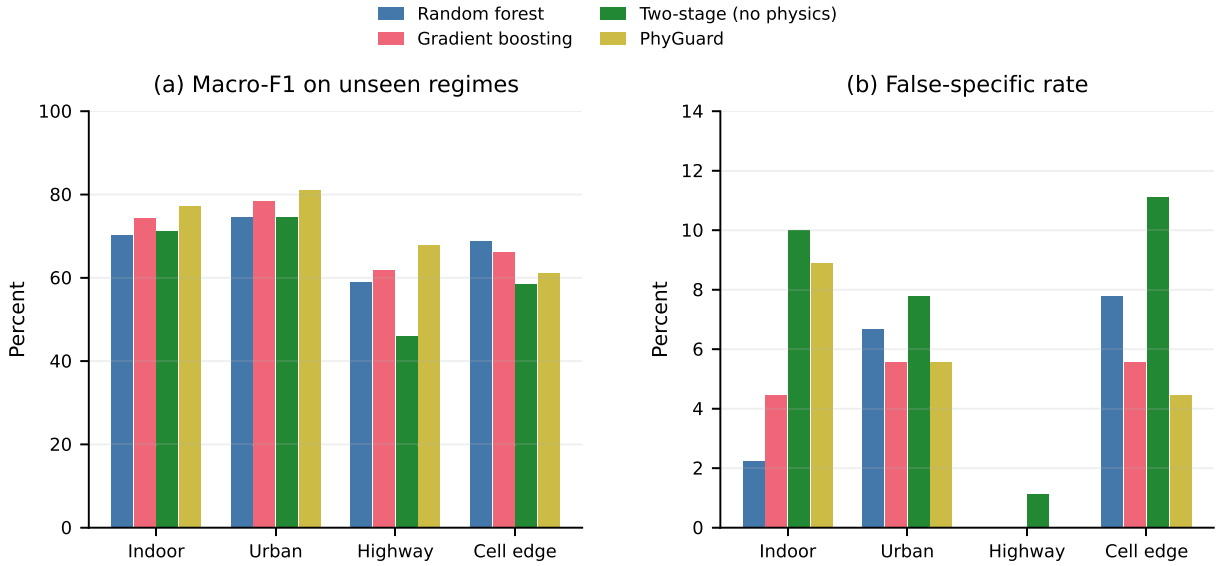


Fig. 7. Performance for each held-out propagation regime. Each regime defines one 270-sequence held-out fold, and the bars report selected accuracy, physical-event coverage, false-specific rate, macro-F1, and balanced accuracy.

6.6. Independently implemented Sionna-based link-level pressure test

Table 7 reports the protocol-frozen locked-interval evaluation. Without Sionna-side training or calibration, the five frozen model repeats achieve 60.41% selected accuracy (correct mechanism among selected physical events), 66.35% physical-event coverage, 39.83% exact physical diagnosis, and 90.08% control abstention. The corresponding 95% sample-cluster bootstrap intervals are [56.74, 64.02], [63.70, 68.84], [36.73, 42.89], and [85.46, 94.26] percentage points. The high control abstention indicates that the selective gate preserves a conservative operating profile under the separately implemented waveform, coding, channel, and receiver stack.

Transfer is mechanism dependent in a physically interpretable manner. Mobility remains highly separable, reaching 95.04% selected accuracy and 80.89% exact diagnosis, a pattern consistent with the persistence of channel-correlation loss and equalization-degradation cues

across generators. Blockage also retains useful transfer, with 66.81% selected accuracy and 54.27% exact diagnosis. Interference and adaptation mismatch are the most sensitive to the separate implementation, indicating where future cross-generator calibration can add the greatest benefit. Across all mechanisms, the 90.08% control-abstention rate indicates that unsupported physical specificity remains strongly suppressed.

6.7. Cross-system directional reliability

Table 8 provides an external directional check of the central SINR-error relation independently of the classifier. The expected negative direction is recovered in the source benchmark, the Sionna benchmark, and the public O-RAN measurement collection. In the source benchmark, 99.13% of eligible sequence-level effects are negative. In Sionna, the aggregate median remains negative despite the separately implemented coded-link stack. In the real-measurement files, the partial Spearman

Table 7

Sionna-based pressure test conditioned on the locked candidate interval. Values are means over five frozen reconstruction repeats. “Decision rate” is physical-event coverage for anomaly rows and abstention for controls; task success is exact mechanism diagnosis for anomalies and abstention for controls.

Task stratum	Decision rate (%)	Selected accuracy (%)	Task success (%)
All physical anomalies	66.35	60.41	39.83
Adaptation mismatch	55.03	22.48	12.37
Blockage	81.24	66.81	54.27
Interference	43.09	23.34	10.06
Mobility	85.11	95.04	80.89
All controls	90.08	–	90.08
Normal control	87.12	–	87.12
Non-physical goodput control	93.00	–	93.00

Table 8

Directional reliability of the SINR–error relation across separately produced systems. Source and Sionna report median within-sequence MCS-stratified Spearman association; the public O-RAN result is the aggregate within-file partial Spearman association controlling continuous MCS.

System	Eligible units	Aggregate association	95% interval	Expected-sign support
Source waveform benchmark	1037 sequences	−0.518	[−0.539, −0.501]	99.13% (1,028/1,037)
Sionna-based benchmark	541 sequences	−0.214	[−0.250, −0.174]	66.17% (358/541)
Public O-RAN measurements	10 files	−0.239	[−0.323, −0.122]	90.00% (9/10 files)

Table 9

Localization quality of the three upstream detectors frozen before formal evaluation. Recall is computed over 706 causal-valid physical anomalies.

Detector	Mean IoU	Recall @0.3 (%)	Recall @0.5 (%)	Mean $ \Delta s $ (frames)
Robust Energy	0.724	78.90	78.05	6.23
PCA Reconstruction	0.745	83.29	82.29	5.35
Isolation Forest	0.668	72.80	71.67	7.65

association is -0.239 after controlling continuous MCS, with a 95% bootstrap interval of $[-0.323, -0.122]$; nine of ten files have the expected sign. Secondary checks also recover negative CQI–BLER association and positive SINR–MCS association after Holm correction.

This cross-system agreement supports the directional physical basis used by PhyGuard while leaving each system free to have its own effect magnitude and calibration.

6.8. Sensitivity to frozen upstream detector proposals

Table 9 reports localization before any detector-conditioned diagnosis is considered. PCA Reconstruction has the highest mean IoU (0.745) and event recall at $\text{IoU} \geq 0.3$ (83.29%). Robust Energy follows with mean IoU 0.724, whereas the source-fitted Isolation Forest reaches 0.668. All three detectors have lower overlap for mild events than for moderate and severe events, showing that weak interventions are more difficult to localize.

Every frozen proposal was propagated through all five downstream model bundles, producing 12,375 detector-conditioned predictions paired with 4125 locked-interval predictions. **Table 10** quantifies the resulting end-to-end operating envelope. PCA Reconstruction provides the strongest detector-conditioned profile, with 50.10% selected accuracy, 31.78% exact diagnosis, and 73.45% control abstention. All three detector definitions and fitting rules were frozen before formal evaluation.

Interval overlap is strongly associated with diagnostic stability. For proposals with $\text{IoU} \geq 0.5$, paired task-success differences relative to the locked interval are only +0.18, −4.61, and −0.51 points for Robust Energy, PCA, and Isolation Forest, respectively. Across detectors, Spearman correlations between IoU and detector-conditioned task success range from 0.429 to 0.548, while correlations between IoU and prediction change range from -0.620 to -0.748 . These results evaluate the intended modular workflow and expose its dependence on localization quality: an external detector proposes an interval, **PhyGuard** extracts

fixed evidence, and the gate–resolver pair emits a mechanism only when the interval supports one.

6.9. Modern temporal-baseline challenge

Table 11 reports all 14 temporal classifiers under the aligned protocol. Under this protocol, PhyGuard obtains $94.32 \pm 1.36\%$ selected accuracy, $83.19 \pm 2.61\%$ physical coverage, $5.56 \pm 1.76\%$ false-specific rate, $78.47 \pm 2.81\%$ exact physical diagnosis, and $94.44 \pm 1.76\%$ control abstention. These values use the locked pipeline and the exact row-level memberships shared by every temporal comparator; **Table 5** reports the separate main-study protocol.

Among models that cover more than 80% of physical events, PhyGuard has the lowest false-specific rate (5.56%) and the highest control abstention (94.44%). MiniRocket, MultiRocket, MultiRocket–HYDRA, and TCN have small positive point differences in selected accuracy, but every paired 95% cluster-bootstrap interval crosses zero. Their false-specific rates exceed PhyGuard by 7.78, 7.22, 8.06, and 8.33 percentage points, respectively; these differences remain supported after Holm adjustment. Consequently, based on the reported mean point estimates, none of the 14 models Pareto-dominates PhyGuard on either the selected-accuracy/coverage/false-specific-rate triplet or the predefined five-metric profile (**Table 12**).

6.10. Physics-formula and validation-policy sensitivity

Across the 12 non-default formula configurations, mean changes relative to the default range from -0.61 to $+1.64$ percentage points for selected accuracy, -1.67 to $+1.81$ for coverage, -1.11 to $+1.94$ for false-specific rate, and -0.07 to $+0.80$ for macro-F1. No tested perturbation causes structural collapse. The similarity-scale sweep is especially stable: selected accuracy remains between 94.04% and 95.83%, coverage

Table 10

Detector-conditioned diagnosis on the Sionna-based set. Deltas are detector-proposed interval minus locked-event interval, in percentage points; brackets give paired 95% sample-cluster bootstrap intervals.

Detector	Selected accuracy (%)	Δ selected accuracy (95% CI)	Exact diagnosis (%)	Δ exact diagnosis (95% CI)
Robust Energy	47.80	−12.62 [−15.41, −9.86]	31.42	−8.41 [−10.58, −6.34]
PCA Reconstruction	50.10	−10.31 [−12.84, −7.82]	31.78	−8.05 [−9.90, −6.26]
Isolation Forest	42.91	−17.51 [−20.51, −14.46]	26.49	−13.34 [−15.74, −10.95]
Detector	Control abstention (%)	Δ control abstention (95% CI)	False-specific rate (%)	Δ false-specific rate (95% CI)
Robust Energy	60.67	−29.41 [−36.79, −22.07]	39.33	+29.41 [+22.07, +36.79]
PCA Reconstruction	73.45	−16.64 [−23.27, −10.10]	26.55	+16.64 [+10.10, +23.27]
Isolation Forest	64.71	−25.38 [−32.71, −18.29]	35.29	+25.38 [+18.29, +32.71]

Table 11

Aligned comparison with all 14 modern temporal classifiers (mean $\pm$ standard deviation over five frozen splits). Higher is better except for false-specific rate.

Method	Selected accuracy (%)	Physical coverage (%)	False-specific rate (%)	Exact physical diagnosis (%)	Control abstention (%)
PhyGuard	94.32 $\pm$ 1.36	83.19 $\pm$ 2.61	5.56 $\pm$ 1.76	78.47 $\pm$ 2.81	94.44 $\pm$ 1.76
ROCKET	93.90 $\pm$ 1.76	93.47 $\pm$ 1.79	10.00 $\pm$ 2.04	87.78 $\pm$ 2.73	90.00 $\pm$ 2.04
MiniRocket	95.33 $\pm$ 1.13	94.86 $\pm$ 1.43	13.33 $\pm$ 2.72	90.42 $\pm$ 0.92	86.67 $\pm$ 2.72
MultiRocket	96.16 $\pm$ 1.28	94.31 $\pm$ 2.12	12.78 $\pm$ 3.97	90.69 $\pm$ 2.73	87.22 $\pm$ 3.97
HYDRA	54.31 $\pm$ 44.37	54.31 $\pm$ 44.35	13.06 $\pm$ 12.03	49.17 $\pm$ 40.19	86.94 $\pm$ 12.03
MultiRocket–HYDRA	96.47 $\pm$ 1.32	94.03 $\pm$ 2.66	13.61 $\pm$ 2.69	90.69 $\pm$ 2.43	86.39 $\pm$ 2.69
Arsenal	94.16 $\pm$ 2.00	92.36 $\pm$ 2.45	10.00 $\pm$ 1.84	86.94 $\pm$ 2.03	90.00 $\pm$ 1.84
FCN1D	94.30 $\pm$ 2.16	86.81 $\pm$ 3.14	11.67 $\pm$ 6.19	81.81 $\pm$ 1.98	88.33 $\pm$ 6.19
ResNet1D	93.30 $\pm$ 2.60	87.22 $\pm$ 1.36	14.17 $\pm$ 3.45	81.39 $\pm$ 3.02	85.83 $\pm$ 3.45
InceptionTime-Lite	93.70 $\pm$ 1.25	95.00 $\pm$ 1.42	9.72 $\pm$ 3.29	89.03 $\pm$ 2.38	90.28 $\pm$ 3.29
LITE1D	93.56 $\pm$ 1.92	81.81 $\pm$ 3.84	11.94 $\pm$ 5.87	76.53 $\pm$ 3.79	88.06 $\pm$ 5.87
TCN	94.47 $\pm$ 1.62	83.47 $\pm$ 4.64	13.89 $\pm$ 3.62	78.89 $\pm$ 5.10	86.11 $\pm$ 3.62
BiGRU	24.00 $\pm$ 22.45	1.25 $\pm$ 1.27	3.06 $\pm$ 2.69	0.42 $\pm$ 0.34	96.94 $\pm$ 2.69
BiLSTM	37.40 $\pm$ 38.39	19.72 $\pm$ 25.19	3.33 $\pm$ 3.24	15.83 $\pm$ 20.77	96.67 $\pm$ 3.24
Tiny Transformer	94.17 $\pm$ 1.14	90.97 $\pm$ 2.28	9.17 $\pm$ 3.24	85.69 $\pm$ 3.06	90.83 $\pm$ 3.24

Table 12

Selected validation-policy operating points at a fixed validation selected-accuracy requirement of 0.90. The validation constraint guides threshold choice but is not a deterministic held-out risk guarantee.

Validation FSR limit	Test selected accuracy (%)	Test coverage (%)	Test FSR (%)
0.025	94.58	77.22	2.78
0.050 (default)	94.32	83.19	5.56
0.100	92.94	87.64	7.78

between 82.22% and 84.03%, and false-specific rate between 5.28% and 5.83%.

Lower adaptation-gap weights increase selected accuracy and control abstention while reducing physical coverage; higher weights move the system toward greater coverage with more false-specific outputs. The MCS-drop penalty shows the same broad operating-profile behavior rather than a discontinuity at $\gamma = 0.35$. These results rule out an isolated fragile operating point within the tested ranges, but they do not establish globally optimal physics weights.

6.11. Packet/block-error indicator ablation

Table 13 reports the frozen per-indicator study. Selected accuracy is pooled over repeat predictions (60.03% for Sionna Full), whereas Table 7 reports the unweighted repeat mean (60.41%); the other rates coincide because their repeat denominators are fixed. All reduced variants meet the prespecified five-point retention margin in both systems. Source selected accuracy remains between 94.77% and 95.84%, indicating that no single evaluated packet/block-error path explains performance. Under Sionna shift, the complete model best balances coverage, exact diagnosis, and control abstention; removing the error-specific score lowers exact diagnosis from 39.83% to 36.18% and control abstention from 90.08% to 82.86%, suggesting partial redundancy with a measurable safety contribution from the complete evidence channel.

6.12. Computational complexity and scalability

Let T denote sequence length, K the number of KPI channels, $d_p = 19$ the physical-feature dimension, $d = 79$ the combined resolver dimension, M the number of Extra-Trees estimators, and D their mean traversal depth. Reference calibration and interval summaries require $O(TK)$ time and $O(K)$ streaming workspace. Directional evidence requires $O(TK + K)$ time. The logistic gate requires $O(d_p)$ time, and Extra-Trees inference requires $O(MD)$ node visits once the fixed-dimensional evidence vector has been extracted (Table 14).

The frozen implementation contains a 19-dimensional logistic gate and a 200-tree Extra-Trees resolver with 34,730 total nodes, mean depth 13.86, maximum depth 18, and a serialized size of 3.42 MB. Under the single-thread environment specified in Section 5, median feature-construction-plus-inference time increases from 3.13 ms at $T = 40$ to 3.21 ms at $T = 80$, 3.30 ms at $T = 320$, and 4.16 ms at $T = 1280$. Thus, a 32-fold increase in sequence length raises median latency by only 1.33 $\times$ in the measured range because fixed-dimensional model inference dominates short windows. Batched inference over prepared evidence vectors scales from 46.8 intervals/s at batch 1 to 5302 intervals/s at batch 128 and 28,843 intervals/s at batch 2,048. Upstream localization and offline training are excluded, and the absolute timings are hardware-specific implementation references.

7. Discussion

7.1. Evidence provided by the combined evaluation

The source-domain study supports three principal conclusions. First, waveform-derived indicators contain complementary information for diagnosing multiple OFDM-link mechanisms without exposing raw IQ samples to the classifier. Second, structured physical evidence is most effective when combined with a learned resolver rather than used as a brittle rule list: relative to the no-physics two-stage model, PhyGuard increases macro-F1 by 10.41 points and coverage by 14.03 points. Third,

Table 13

Per-indicator ablation under the aligned source and Sionna-based protocols. Values are percentages pooled over five frozen repeats.

System	Variant	Selected accuracy	Coverage	False-specific rate	Exact diagnosis	Control abstention
Source	Full	94.32	83.19	5.56	78.47	94.44
	No raw error statistics	94.77	82.36	5.28	78.06	94.72
	No error physics score	95.50	83.33	6.11	79.58	93.89
	No error-indicator path	95.84	83.47	5.56	80.00	94.44
Sionna	Full	60.03	66.35	9.92	39.83	90.08
	No raw error statistics	60.19	62.97	10.76	37.90	89.24
	No error physics score	58.52	61.81	17.14	36.18	82.86
	No error-indicator path	60.30	59.15	14.79	35.67	85.21

Table 14

Online complexity of the frozen PhyGuard pipeline.

Stage	Time	Working memory
Reference calibration and summaries	$O(TK)$	$O(K)$
Physical-consistency evidence	$O(TK + K)$	$O(K + d_p)$
Logistic evidence gate	$O(d_p)$	$O(d_p)$
Extra-Trees resolver	$O(MD)$	$O(d)$

abstention is operationally essential; the forced-output variant labels every control as a physical fault, whereas the selective system reduces the false-specific rate to 5.28%.

The Sionna-based benchmark extends this evidence beyond the source generator. Without Sionna-side fitting, the gate abstains on 90.08% of controls, mobility reaches 80.89% exact diagnosis, and blockage reaches 54.27%. The cross-system reliability analysis further shows that the expected SINR–error direction is present in the source benchmark, the separately implemented coded-link benchmark, and ten public O-RAN measurement files. These results support the intended role of physics as an inductive structure whose expected direction persists across the evaluated assets rather than as a generator-specific label encoding.

The temporal challenge clarifies the value of the selective objective. Several high-capacity temporal models increase coverage, but among models with more than 80% coverage PhyGuard has the lowest false-specific rate and highest control abstention. Paired analysis does not resolve the small selected-accuracy differences in favor of the strongest random-kernel models, while their false-specific penalties remain statistically supported. Based on the reported mean point estimates, PhyGuard therefore provides a non-dominated operating profile that prioritizes reliable mechanism emission rather than unconditional classification.

The formula, policy, and indicator studies show that this profile is not tied to one fragile numerical setting. Reasonable formula perturbations move selected accuracy, coverage, and false-specific rate within narrow ranges; the policy grid exposes a controllable safety–coverage continuum; and all packet/block-error ablations retain selected accuracy within the prespecified margin. The complete error-indicator path nevertheless gives the strongest Sionna balance, illustrating how redundant evidence can still improve robustness under shift.

Beyond the wireless instantiation, the methodological contribution is the separation of evidence sufficiency from class attribution. This provides an auditable alternative to confidence-only rejection when domain relations can reveal contradictory or non-specific evidence. The present experiments evaluate this principle for OFDM telemetry; transfer to other systems would require domain-specific evidence definitions and independent validation.

7.2. Operational interpretation and scope

PhyGuard is designed as a modular post-alarm interface. An external detector supplies a candidate interval and synchronized telemetry; PhyGuard then computes fixed reference-calibrated evidence and either emits a mechanism-specific diagnosis or abstains. The three-detector

evaluation assesses this interface under three frozen proposal mechanisms. When detector overlap reaches $\text{IoU} \geq 0.5$, downstream task success remains close to the locked-interval reference, highlighting localization quality as a practical integration requirement rather than a change to the diagnostic model itself.

The current evidence is link-level rather than over-the-air deployment validation. The task targets one dominant mechanism per interval, assumes an available normal reference prefix, and evaluates 1×2 SIMO CDL-A/B/C in the Sionna-based benchmark. The source PER is a compact uncoded reliability surrogate, whereas the Sionna analysis supplies coded-link BLER evidence. Future work can extend the same selective interface to mixed-fault targets, broader antenna/channel configurations, persistent-fault baselines, and detector-aware calibration. These boundaries preserve a clear scope while leaving the proposed architecture and operating principle unchanged.

8. Conclusion

This paper presented PhyGuard, a physics-guided selective diagnostic interface for multivariate OFDM telemetry. On the 1,080-sequence waveform-grounded benchmark, PhyGuard achieves the best observed macro-F1 and physical-event coverage among the selective tabular methods while maintaining a 5.28% false-specific rate. The learned gate and resolver convert auditable communication-domain relations into a controllable coverage–risk operating point rather than forcing a mechanism label for every alarm.

An independently implemented, protocol-frozen Sionna-based benchmark, three frozen upstream detectors, 14 temporal classifiers, fixed-model sensitivity sweeps, cross-system reliability analysis, per-indicator ablation, and empirical scaling substantially broaden the evaluation. Based on the reported mean point estimates, no temporal baseline Pareto-dominates PhyGuard across the predefined selective metrics; among high-coverage models, PhyGuard provides the lowest false-specific rate. The Sionna-based benchmark preserves 90.08% control abstention without Sionna-side fitting, while cross-system analysis recovers the expected SINR–error direction in source, Sionna, and public O-RAN measurements. Indicator ablation suggests that performance is distributed across partially redundant evidence paths, while the complete model provides the strongest shifted-domain safety balance. Together, these results support PhyGuard as an efficient, auditable, and risk-controlled post-alarm diagnostic framework for OFDM-link monitoring.

CRedit authorship contribution statement

Chunyan Wang: Writing – review & editing, Supervision, Project administration; **Yiming Sun:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis.

Ethics declaration

The author(s) declare(s) that the study does not involve humans nor animal subjects.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) for language refinement and software-debugging support. The authors reviewed and edited all generated material and independently executed and verified the simulations, statistical analyses, and reported results, taking full responsibility for the content.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The generated data and implementation are not deposited in a public repository. Derived result tables, protocol summaries, and quality-control manifests are available from the corresponding author upon reasonable request. No personal or proprietary data are used.

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